Until we are all equal

Participatory Climate Risk and Vulnerability Assessment for Gender Responsive Climate Adaptation and Preparedness Plans (GRAPP) in Kwale County.

Final Kwale County Community Risk Preparedness Plan 2025-2030

Consultant

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List of Acronyms

AAR	After Action Review
CADP	County Annual Development Plan
CCCCF	County Climate Change Fund
CCCSC	County Climate Change Steering Committee
CDCC	County Disaster Command Centre
CDMU	County Disaster Management Unit
CEOP	County Emergency Operations Protocol
CECM	County Executive Committee Member
CIDP	County Integrated Development Plan
CLU	County Learning Unit (Knowledge Hub)
CBO	Community-Based Organization
CDMC	County Disaster Management Committee
CEC	County Executive Committee
CFAs	Community Forest Associations
CSO	Civil Society Organization
CSR	Corporate Social Responsibility
DRM	Disaster Risk Management
DRR	Disaster Risk Reduction
EWS	Early Warning System
FAO	Food and Agriculture Organization
GBV	Gender Based Violence GIS Geographic Information System
IPCC	Intergovernmental Panel on Climate Change
IRI	Integrated Resource Inventory
KCCAP	Kenya Climate Change Action Plan
KFS	Kenya Forest Service
KLH	Knowledge and Learning Hub
KMD	Kenya Meteorological Department
KMFRI	Kenya Marine and Fisheries Research Institute
KRCS	Kenya Red Cross Society
KSH	Kenya Shilling
KYEOP	Kenya Youth Employment and Opportunities Project
MSCP	Multi-Stakeholder Coordination Platform
NDMA	National Drought Management Authority
NDC	Nationally Determined Contribution
NDOC	National Disaster Operations Centre
NEMA	National Environment Management Authority
NFIs	Non-Food Items
NGO	Non-Governmental Organization
NRM	Natural Resource Management
PDAR	Post-Disaster Assessment Report
PFMA	Public Finance Management Act
PIFS	Public Information and Feedback System
PCVA	Participatory Climate Vulnerability and Capacity Assessment
PADs	Persons Abled Differently
PPP	Public–Private Partnership
RMPU	Resource Mobilization and Partnerships Unit
SACCO	Savings and Credit Cooperative Organization
SDGs	Sustainable Development Goals
SRHR	Sexual and Reproductive Health and Rights
SSP	Shared Socioeconomic Pathway
UNDP	United Nations Development Programme

VSLA	Village Savings and Loan Association
VDRT	Village Disaster Response Team
WCCPC	Ward Climate Change Planning Committee
WCF	Ward Climate Fund
WEF	Women Enterprise Fund
WRUA	Water Resource Users Association
WWF	World Wide Fund for Nature

1 Introduction

1.1 Purpose

The Kwale County Community Risk Preparedness Plan (2025–2030) provides a strategic framework to enhance the county's capacity to anticipate, prepare for, respond to, and recover from the impacts of climate-induced hazards. The plan is designed to safeguard human lives, livelihoods, critical infrastructure, and ecosystems while promoting adaptive development pathways in line with the county's long-term resilience agenda. It responds to the increasing frequency and severity of droughts, floods, coastal erosion, pest and disease outbreaks, and strong winds that threaten both rural and urban communities in Kwale.

The purpose of this plan is to reduce vulnerability to climate and disaster risks through proactive preparedness measures that strengthen institutional coordination, community awareness, and local response systems. It seeks to ensure that essential services and socio-economic activities within the county remain functional even under extreme events, and that recovery processes are rapid, inclusive, and sustainable. The plan also complements the implementation of the Kwale County Climate Change Action Plan (2023–2027), the County Disaster Risk Management Framework, and Kenya's National Adaptation Plan (2015–2030) by translating national and county-level commitments into ward-level action.

Importantly, the plan recognizes that the effects of climate change are not gender-neutral. Women, youth, and persons abled differently (PADs) often face greater exposure and fewer resources to recover from disasters. For instance, women are typically responsible for household water collection and fuelwood gathering, activities that become more difficult during drought or flooding. Therefore, this plan mainstreams gender equality and social inclusion across all preparedness components, ensuring equitable access to information, resources, and decision-making platforms.

Through multi-sectoral coordination, community engagement, and evidence-based planning, this document provides a blueprint for building a climate-resilient Kwale County. It integrates lessons from Participatory Climate Vulnerability and Capacity Assessments (PCVAs) conducted in Vanga, Ramisi, and Pongwe Wards, which revealed the unique exposure patterns, coping strategies, and local adaptation practices across coastal, agricultural, and semi-arid landscapes of the county.

1.2 Scope

The scope of the Kwale County Community Risk Preparedness Plan extends across all four sub-counties ; Matuga, Msambweni, Kinango and Lunga Lunga, while giving detailed focus to the coastal wards of Vanga, Ramisi, and Pongwe where participatory assessments were conducted. It encompasses a five-year period (2025–2030) and integrates multiple sectors including water, agriculture, fisheries, infrastructure, health, and environment.

This plan targets both institutional systems (county departments, emergency units, NGOs, private sector actors) and community-level structures (Ward Climate Change Planning Committees, Village Disaster Response Teams, and local groups such as BMUs, WRUAs, and CFAs). By doing so, it ensures vertical coordination between local and county-level governance and horizontal integration across thematic sectors.

Geographically, Kwale County spans approximately 8,270 km², characterized by coastal lowlands, rolling uplands, and sections of semi-arid hinterland. It borders the Indian Ocean to the east, making it prone to coastal hazards such as sea-level rise, storm surges, and saltwater intrusion. Inland zones are affected by droughts, flash floods, and pest infestations, while deforestation and land degradation exacerbate vulnerability.

The plan also covers both rapid-onset hazards (e.g., floods, strong winds, pest outbreaks) and slow-onset hazards (e.g., drought, environmental degradation, coastal erosion). It provides mechanisms for prevention, mitigation, preparedness, response, and recovery, ensuring continuity of operations in critical sectors.

Special emphasis is given to gender-responsive and socially inclusive approaches to ensure that women, youth, and PADs are not only beneficiaries but also active participants in risk preparedness and response planning.

2 Climate Risk Assessment

2.1 Hazard Identification

Kwale County experiences a diverse range of climate-induced hazards due to its geographical positioning along Kenya's coastal belt and its topographic variation from the humid coastal plains to the semi-arid hinterlands. The major hazards identified through participatory climate vulnerability and capacity assessments (PCVAs) conducted in Vanga, Ramisi, and Pongwe Wards (2025) and corroborated by the Kwale County Hazard Risk and Vulnerability Profile (NDMA, 2022), include drought, floods, pests and diseases, environmental degradation, human–wildlife conflicts, and strong winds (Table 1).

Drought is the most recurrent hazard, affecting nearly all wards in the county. Extended dry periods have become more frequent and severe, resulting in reduced water availability, declining agricultural productivity, and increased competition over natural resources. In the inland areas of Kinango, Samburu, and Shimba Hills sub-counties, drought triggers food insecurity and livestock losses. Along the coast, communities in Vanga, Ramisi, and Pongwe experience saline water intrusion and reduced freshwater flows into estuarine and mangrove systems.

Flooding, on the other hand, remains a cyclical challenge particularly during the long rains (March–May). Low-lying areas such as Ramisi, parts of Lunga Lunga, and Shimba Hills receive excessive runoff that inundates farms and settlements. Floods cause damage to roads, bridges, and public facilities such as schools and dispensaries, and disrupt learning and healthcare delivery. In Vanga, periodic riverine floods from the Umba River and storm surges along the Indian Ocean coastline compound the community's vulnerability.

Pest and disease outbreaks are emerging as secondary hazards linked to changing temperature and rainfall patterns. The proliferation of crop pests such as the fall armyworm (Spodoptera frugiperda) and livestock diseases like East Coast fever have led to recurrent losses among smallholder farmers and pastoralists. Vector-borne human diseases including malaria and dengue fever are also on the rise, particularly after floods.

Environmental degradation, manifested through deforestation, sand harvesting, charcoal burning, and mangrove destruction, is an underlying hazard that heightens vulnerability. The reduction of vegetative cover accelerates soil erosion, reduces infiltration, and alters local microclimates. Meanwhile, human–wildlife conflicts have escalated around Shimba Hills and Mwalewa areas where elephants and baboons invade farms, destroying crops and threatening human safety.

Lastly, strong winds, often associated with monsoon transitions and tropical storms, pose risks to coastal settlements, fishing communities, and nearshore infrastructure. Destruction of housing, capsizing of fishing vessels, and damage to power lines and communication networks are common.

Table 1: Key Climate Hazards in Kwale *County (*Source: Kwale County Hazard Risk and Vulnerability Profile (NDMA, 2022 and PCVA 2025)

Hazard	Frequency	Most Affected Areas	Key Impacts
Drought	High (annual to biannual)	Kinango, Shimba Hills, Pongwe, Ramisi	Crop failure, water scarcity, livestock mortality
Floods	Moderate to High	Ramisi, Vanga, Lunga Lunga	Infrastructure damage, displacement, disease outbreaks
Pests & Diseases	Moderate	County-wide	Crop and livestock losses, human health risks
Environmental Degradation	High	Shimba Hills, Kinango, Vanga	Soil erosion, loss of biodiversity, reduced ecosystem services
Human–Wildlife Conflicts	Moderate	Shimba Hills, Mwalewa	Crop destruction, injuries, loss of income
Strong Winds	Seasonal	Coastal areas (Vanga, Ramisi, Pongwe)	Damage to housing, marine accidents

2.2 Vulnerability Analysis

Vulnerability to climate hazards in Kwale County is driven by a complex interplay of environmental, socio-economic, and institutional factors. The PCVAs conducted across Vanga, Ramisi, and Pongwe Wards revealed that livelihood dependence on natural resources, particularly fishing, farming, and forestry, amplifies exposure to climatic variability (Table 2).

In Vanga, households depend heavily on fishing and mangrove harvesting. Disruptions caused by storms or declining fish stocks directly affect household income and food security. In Ramisi, vulnerability stems from monoculture sugarcane farming and poor drainage systems that exacerbate flood damage. Pongwe faces severe water scarcity and degraded soils, limiting agricultural resilience.

From a gender and inclusion perspective, women are disproportionately affected due to their socio-economic roles and limited access to productive resources. They often bear the brunt of unpaid labour during hazards, walking longer distances for water, caring for the sick, and managing food shortages. In addition, a United Nations (UN) Study (Climate crisis driving surge in gender-based violence, UN study finds | UN News) findings demonstrate that extreme weather, displacement, food insecurity, and economic instability are key factors increasing the prevalence and severity of gender-based violence. These impacts hit hardest in fragile communities, where women already face entrenched inequalities and are more vulnerable to assault. According to the study every 1°C rise in global temperature is associated with a 4.7 per cent increase in intimate partner violence (IPV), the study cites. In a 2°C warming scenario, 40 million more women and girls are likely to experience IPV each year by 2090. In a 3.5°C scenario, that number more than doubles. Men, particularly those involved in fishing and charcoal production, are at high risk of occupational hazards and economic displacement.

Youth unemployment further compounds vulnerability, leading to migration and reduced community cohesion. Persons abled differently (PADs) face barriers to participation in adaptation and emergency response processes due to inadequate infrastructure, communication challenges, and social marginalization. Elderly populations, particularly widows and those living alone, are also less mobile during evacuations and often excluded from decision-making forums.

Institutionally, limited interdepartmental coordination, inadequate early warning dissemination, and constrained financial resources weaken the county's adaptive capacity. The

absence of clear linkages between county departments, ward committees, and national agencies like the National Drought Management Authority (NDMA) slows down emergency response.

Table 2: Vulnerability Context in Kwale County (Source: PCVA,2025)

Vulnerable Group	Source of Vulnerability	Climate Impact	Required Support
Women	Surging GBV cases, underrepresentation in natural resource management & climate adaptation planning	Food insecurity, energy and water stress	Gender-sensitive resource access, empowerment programs
Youth	Unemployment, limited participation in governance	Loss of income opportunities	Skills development, vocational training
PADs	Inaccessibility to services and information	Isolation during emergencies	Inclusive infrastructure, targeted communication
Elderly	Mobility constraints, dependency and underrepresentation in decision making	Delayed evacuation, health complications	Targeted assistance, social protection as well as mainstreaming in adaptation decision making
Fisherfolk	Marine resource decline, storm surges	Loss of livelihood assets	Weather alerts, diversified income sources

2.3 Impact Evaluation

The cumulative impacts of climate hazards on Kwale County are extensive, affecting both the natural environment and human well-being. Economic losses are estimated at KES 1.8 billion annually (NDMA, 2023), mainly from crop failure, damaged infrastructure, and disrupted fisheries.

During prolonged droughts, surface water sources such as pans, wells, and rivers dry up, forcing women and children to trek over 5–8 km daily to fetch water. This increases the risk of gender-based violence (GBV) and limits time for income-generating activities. Food security declines sharply, with malnutrition rates rising among children and expectant mothers.

Floods contribute to disease outbreaks such as malaria, cholera, and diarrhoea. In 2022, approximately 12,000 people were displaced in Msambweni and Lunga Lunga sub-counties due to flash floods (County Disaster Unit, 2023). Infrastructure repair costs, including roads and bridges, averaged KES 300 million per flood event.

Ecosystem impacts are equally significant. Deforestation around Shimba Hills reduces carbon sequestration potential, while mangrove loss in Vanga and Ramisi weakens coastal protection and nursery habitats for fish. Overexploitation of land and water resources has degraded soil fertility, reduced groundwater recharge, and heightened conflict over natural resources.

Climate risks also threaten the continuity of essential services. Health facilities experience medicine stockouts during floods, schools close for extended periods, and markets remain inaccessible due to impassable roads. The cumulative social burden is felt most acutely by low-income households that lack savings or insurance mechanisms.

These impacts underline the urgent need for integrated preparedness strategies that combine infrastructure resilience, ecosystem restoration, social protection, and early warning systems.

3 Objectives

3.1 Preparedness Goals

The overarching goal of the Plan is to strengthen the county's resilience to climate-induced hazards by reducing exposure, minimizing vulnerability, and enhancing adaptive capacity across all sectors. Specifically, the plan aims to protect lives, safeguard critical infrastructure, secure livelihoods, and ensure the continuity of essential services during and after disaster events.

Over the next five years, the county seeks to operationalize a proactive disaster risk management system that integrates early warning, community preparedness, and rapid response mechanisms. These efforts will be anchored on the principles of inclusivity, sustainability, and accountability, ensuring that interventions benefit all residents, particularly women, youth, and persons abled differently (PADs), who are often disproportionately affected by disasters.

To achieve these aims, the county government, in collaboration with national agencies and development partners, will focus on the following key goals:

- Enhancing institutional coordination for climate risk management across departments and administrative levels.
- Protecting critical infrastructure such as schools, health facilities, roads, and water systems from hazard impacts.
- Safeguarding vulnerable populations by establishing social safety nets, cash transfer programs, and ward emergency funds.
- Promoting resilient livelihoods through climate-smart agriculture through training on resilient design approach from homestead to watershed level including climate-smart agriculture, sustainable fisheries, eco-enterprises focusing on circular economies, and restoration of degraded ecosystems.
- , sustainable fisheries, and ecosystem restoration.
- Improving early warning and information systems to ensure timely dissemination of risk alerts to all communities, including those in remote areas.
- Mainstreaming gender equality and social inclusion in all preparedness, response, and recovery actions.

3.2 Success Metrics

To monitor progress toward the achievement of these goals, specific success metrics have been developed to guide implementation and evaluation. These indicators are both quantitative and qualitative, enabling assessment of outcomes related to institutional performance, community readiness, and gender equity (Table 3).

- **The key performance metrics include:**Targeted GBV anticipatory actions for drought, floods and strong winds climate hazards
- Gender-responsive governance- inclusion of women, youth and PADs through capacity building across all villages
- Reduction in disaster-related losses: measurable decrease in the number of households displaced, lives lost, and assets damaged by floods and droughts.
- Improved response time: reduced average time taken to mobilize emergency teams and distribute relief supplies following hazard onset.
- Increased community awareness: higher percentage of households trained on resilience design from homestead to watershed level, early warning signs, emergency procedures, and adaptive practices.
- Enhanced livelihood resilience: increased adoption of drought-tolerant crops, improved post-harvest management, and diversification of income sources among households.

- Gender inclusion: higher representation of women and PADs in Ward Climate Change Planning Committees (WCCPCs), Community Forest Associations (CFAs), Beach Management Units (BMUs), Water Resource Users Associations (WRUAs) and Village Disaster Response Teams (VDRTs).
- Strengthened institutional capacity: establishment and operationalization of a County Disaster Risk Management Unit with clear coordination mandates.
- Ecosystem restoration outcomes: hectares of degraded forest, mangrove, and watershed areas rehabilitated under community stewardship.

Table 3: Sample preparedness Success Metrics for Kwale County (Source: PCVA,2025)

Objective Area	Baseline (2025)	Target (2030)	Key Indicator
Community disaster awareness	40% of households aware	80%	Percentage of population trained in early warning and response
Women in leadership in Disaster Risk Reduction	25%	50%	Representation of women in Ward Committees and County DRM Unit
Functional early warning systems	2 sub-counties	4 sub-counties	Proportion of wards with operational community EWS
Resilient livelihoods	30% households diversified	70%	Adoption of climate-smart farming and alternative income activities
Ecosystem restoration	2,000 ha restored	6,000 ha	Area of rehabilitated forest, mangrove, and watershed ecosystems
Response time to hazards	72 hours	24 hours	Average time between hazard alert and emergency action

4 Preparedness Strategies

4.1 Prevention and Mitigation

Prevention and mitigation constitute the foundation of Kwale County's climate risk management strategy. The goal is to reduce the likelihood of disasters occurring and minimize their adverse effects through structural and non-structural measures. Given the increasing intensity of droughts, floods, and environmental degradation, the county has prioritized integrated land and water resource management, resilient infrastructure development, and nature-based solutions (NbS).

One of the key prevention actions involves upgrading critical infrastructure to withstand climate stressors. The Department of Roads, Public Works, and Infrastructure, in collaboration with the Kenya Rural Roads Authority (KeRRA) and the County Disaster Management Unit, will strengthen drainage systems in flood-prone zones such as Ramisi, Kikoneni, and Lunga Lunga. This will include desilting culverts, constructing stormwater channels, and reinforcing bridge foundations. Schools and dispensaries located in low-lying areas will be elevated or flood-proofed to ensure continuity of learning and healthcare services. In areas where persistency and adaptation measures are not viable for schools and health care facilities, these may need to be relocated considering accessibility by the local community within the context of long-term adaptation to eliminate exposure to flooding.

In drought-prone areas like Pongwe and Kinango, water harvesting and storage structures, such as solar-powered boreholes, rock catchments, and elevated storage tanks, will be expanded to support domestic and livestock needs. The county will also promote drought-tolerant crop varieties (e.g., *Sorghum bicolor* and *Vigna unguiculata*) and agroforestry practices integrating Neem (*Azadirachta indica*) and Casuarina (*Casuarina equisetifolia*) species to enhance soil fertility and prevent erosion.

Ecosystem-based mitigation will focus on restoring degraded mangrove forests in Vanga and Ramisi, rehabilitating catchment areas in Shimba Hills, and reforesting communal lands. These actions not only stabilize shorelines and improve biodiversity but also provide carbon sequestration benefits aligned with Kenya's Nationally Determined Contributions (NDCs).

Non-structural measures will include enforcing land-use planning regulations, restricting settlement in flood plains, and mainstreaming environmental safeguards in development approvals. Awareness campaigns on sustainable resource use, afforestation, and waste management will be carried out through community barazas, radio programs, and school climate clubs.

The integration of gender-sensitive mitigation will ensure that women and PADs participate actively in afforestation projects, environmental committees, and water management groups, recognizing their traditional knowledge and custodianship of natural resources. This will be done through consulting them on the labour inputs that would best serve them and the regenerative projects they are involved in including sharing unpaid care work with men.

4.2 Preparedness

Preparedness focuses on strengthening institutional readiness, community awareness, and logistical systems to enable timely and coordinated response to disasters. The county's preparedness framework is built on capacity-building, early warning systems (EWS), contingency planning, and resource pre-positioning (Table 4).

To enhance community readiness, Ward Climate Change Planning Committees (WCCPCs) and Village Disaster Response Teams (VDRTs) in Vanga, Ramisi, and Pongwe will be trained in risk

mapping, emergency response, and first aid. Simulation exercises and evacuation drills will be organized at least twice annually, led by the County Disaster Management Unit (CDMU) with support from the Kenya Red Cross Society (KRCS), NDMA, and Plan International.

The county will strengthen its multi-hazard early warning system, integrating meteorological data from the Kenya Meteorological Department (KMD) with local indigenous knowledge. Community radio broadcasts, SMS alerts, and digital platforms will be used to disseminate warnings in real time, ensuring accessibility to women, PADs, and those in remote areas. Among the digital platforms, the Kenya Digital Agriculture Platform run by Kenya Agricultural and Livestock Research Organization (KARLO) would particularly be useful as it delivers multiple benefits incorporating weather forecasts, agronomic advisory, agricultural insights and market information at the ward level. This would particularly be useful in climate smart agriculture where early planting and use of quality seeds becomes an adaptation strategy to optimize harvests against rainfall variability.

Health preparedness will involve stockpiling essential medicines, strengthening surveillance for vector-borne diseases, and maintaining functional cold chains for vaccines. More importantly, the health preparedness will encompass GBV response by providing first aid and treatment services and testing to aid in investigations and prosecutions. The health and education centres should also provide Sexual and Reproductive Health and Rights awareness, sensitization and accessibility to information and supplies. Education preparedness will focus on incorporating disaster awareness into school curricula and establishing safety clubs in collaboration with the Ministry of Education.

Each ward will maintain emergency storage sites stocked with essential supplies such as food, water, tents, dignity kits, first-aid and rape kits as well as SRHR supplies. Special provisions will be made for sanitary products and assistive devices for PADs.

Institutionally, the county will develop ward-level contingency plans that clearly outline responsibilities, resource requirements, and communication protocols. The CDMU will maintain a countywide emergency contact database for rapid mobilization.

Table 4: Key Preparedness Interventions (2025–2030)

Intervention	Responsible Agency	Coverage Area	Expected Outcome
Community early warning system	NDMA, KMD, County DRM Unit, KARLO	4 sub-counties	Timely hazard alerts reaching at least 90% of residents
Emergency drills and training	CDMU, KRCS, Plan International	All wards	Enhanced community readiness and reduced response time
Ward contingency plans	County Planning Department	20 wards	Functional ward-level preparedness mechanisms
Health and sanitation preparedness	County Health Department	Flood-prone zones	Reduced disease outbreaks during emergencies
Inclusive education and awareness	County Education Department	Schools and youth groups	Increased awareness and participation of women and youth

4.3 Response

The response strategy aims to ensure rapid, coordinated, and equitable action immediately following the occurrence of a disaster. Effective response relies on clear communication, pre-assigned roles, and readily available emergency resources.

When a hazard event occurs, the County Disaster Command Centre (CDCC), under the Office of the Governor, will activate the County Emergency Operations Protocol (CEOP). The CEOP will coordinate with the NDMA, the National Police Service, Kenya Red Cross, and sectoral departments to assess damage, mobilize rescue operations, and deliver humanitarian support.

Evacuation procedures will prioritize women, children, PADs, and the elderly, ensuring their safe relocation to designated shelters equipped with sanitation facilities, privacy zones, and medical support. The county will establish ward-level safe havens (e.g., school compounds and churches) that double as evacuation centers. Each center will maintain an emergency stockpile of water, food rations, bedding, and first-aid supplies.

A Communication and Command Structure will guide decision-making, with the Governor serving as the County Disaster Incident Commander. Sub-county administrators will lead localized response teams, supported by community volunteers and local NGOs. Real-time information sharing through radio, mobile phones, and WhatsApp coordination groups will ensure that updates reach all actors swiftly.

A gender desk will be established within the CDMU to coordinate relief efforts that specifically address the needs of women and vulnerable groups. This will include access to dignity kits, psychosocial support, and protection from gender-based violence (GBV) in evacuation centers and continuity of SRHR services.
Monitoring of response effectiveness will be continuous, guided by post-incident debrief sessions and field verification of relief distribution.

4.4 Recovery

The recovery phase focuses on restoring normalcy, rebuilding resilience, and reducing future vulnerability. Recovery efforts will be both short-term, addressing immediate humanitarian needs, and long-term, focusing on reconstruction and livelihood diversification.

In the immediate aftermath of a disaster, the County Government of Kwale, supported by development partners, will provide emergency cash transfers to affected households, particularly those led by women, the elderly, and PADs. In addition, the county will put in place anticipatory actions to prevent and address GBV cases which have been shown to increase in tandem with disaster occurrences. Livestock restocking, seed distribution, and rehabilitation of damaged infrastructure will follow.

Long-term recovery initiatives will prioritize the "build back better" approach, integrating climate resilience into reconstruction. For example, roads and bridges washed away by floods will be redesigned with adequate culverts, elevated embankments, and reinforced concrete. Schools and health centers will be retrofitted to withstand extreme weather.

Livelihood restoration will focus on supporting microenterprises, promoting savings and credit cooperatives (SACCOs), and linking vulnerable households to economic empowerment programs such as the Kenya Youth Employment and Opportunities Project (KYEOP) and Women Enterprise Fund (WEF). Plan International is supporting and building the capacity of Village Savings and Loan Associations (VSLAs) in Kwale County targeting women groups. The method is a life-changing approach, especially for the community members living in poverty

with limited economic capacity, to engage in productive businesses and to secure an income. In Kenya, the transformation applying VSLAs among members has been significant in enhancing resilience especially for rural and indigenous women. In addition, VSLAs are vehicles of inclusion, providing women and young women and men with opportunities to access capital to invest in their economic endeavors whether it be profiting businesses or self-employment. The method provides a much-needed solution to addressing barriers to access financial services such as lack of collateral, stringent terms and conditions from banks, physical access to financial institutions, prohibitive interest rates and minimum transaction amounts. Barriers that keep people living in poverty in the margin. Moreover, VSLAs provide accessible source of capital for women and other marginalized groups to more than to start up and run their own businesses, but also to provide for family needs as well as offering women an opportunity to exercise leadership and decision making within households, cooperatives, and communities. The VSLAs are self-managed and have encouraged women to step up and take leadership roles that often have led to other aspects than ensuring economic autonomy for the women, such as strengthened communities. Ecosystem recovery will include rehabilitation of degraded mangrove and forest zones, on farm forest production systems, and water catchment areas. These restoration projects will also serve as income-generating activities for women and youth groups through sell of tree and fruit seedlings, planting initiatives and ultimate harvests of on farm forest products for timber and wood fuel. In addition, restoration activities could include nature-based enterprises such as beekeeping delivering multiple benefits of conservation and livelihoods diversification to combat vulnerability and enhance climate change adaptation and preparedness. The restoration activities should be aligned with both Kenya's Forest and Landscape Restoration Plan (FOLAREP) 2023-2027 and the National Landscape and Ecosystem Restoration Strategy 2023-2032 (Towards Growing 1.5 billion Trees Initiative).

The CDMU, in collaboration with NDMA and the County Treasury, will establish a Post-Disaster Needs Assessment (PDNA) mechanism to quantify losses and guide budget reallocation for recovery programs.

5 Roles and Responsibilities

5.1 Climate Risk Management Team

The effective implementation of the Kwale County Community Risk Preparedness Plan (2025–2030) depends on clear institutional coordination, role delineation, and collaboration among all key actors. The County Government of Kwale will provide overall leadership through the County Disaster Management Unit (CDMU) housed under the Department of Environment, Climate Change, and Natural Resources. This unit will serve as the coordinating entity responsible for developing, operationalizing, and monitoring preparedness, response, and recovery activities across all sectors.

The Governor's Office will oversee strategic decision-making and resource mobilization, ensuring that climate preparedness remains a cross-cutting development priority. The County Executive Committee Member (CECM) for Environment and Climate Change will chair the County Climate Change Steering Committee (CCCSC), which brings together representatives from line departments, civil society, and private sector partners.

Each sectoral department will play a defined role in preparedness planning. The Department of Water will focus on drought preparedness through the development and maintenance of water infrastructure, while the Department of Agriculture will promote climate-smart practices and pest surveillance. The Department of Health will coordinate disease surveillance and emergency medical response, and the Department of Infrastructure will lead in strengthening roads and flood protection works.

At the community level, Ward Climate Change Planning Committees (WCCPCs) and Village Disaster Response Teams (VDRTs) will serve as the primary structures for mobilization, early warning dissemination, and first-line response. These teams will maintain linkages with the county units and report incidents through established communication channels.

Non-governmental organizations (NGOs) and civil society organizations (CSOs) such as Plan International, WWF-Kenya, and Kenya Red Cross Society (KRCS) will complement government efforts by supporting community capacity-building, gender mainstreaming, and humanitarian response. Private sector entities will be encouraged to invest in local resilience initiatives as part of their corporate social responsibility (CSR) programs.

National agencies such as the National Drought Management Authority (NDMA), Kenya Meteorological Department (KMD), and National Environment Management Authority (NEMA) will provide technical backstopping, policy guidance, and monitoring support to align county efforts with national priorities.

Table 5: Institutional Roles in Climate Risk Preparedness

Institution	Key Responsibilities	Coordination Mechanism
Office of the Governor	Policy oversight, emergency declarations, resource mobilization	County Steering Committee
Department of Environment and Climate Change, Gender, SRHR and PADs	Lead coordination, policy implementation, reporting	County Disaster Management Unit
NDMA	Drought early warning, vulnerability assessment, response coordination	County EWS network
KMD	Weather forecasting, climate data sharing	County information hub
KARLO	Conduct agricultural research, develop and promote climate-	Agricultural Research and Extension Liaison Platform

Institution	Key Responsibilities	Coordination Mechanism
	resilient crop and livestock varieties, soil fertility management, and adaptive farming technologies	
County Department of Agriculture	Climate-smart farming, pest and disease control	Agricultural subcommittee
County Department of Water	Water infrastructure, drought mitigation	Water Sector Working Group
County Department of Health	Emergency response, disease surveillance	Health Disaster Desk
County Department of Infrastructure	Road maintenance, flood control	Public Works Coordination Forum
Ward Climate Change Planning Committees	Local coordination, awareness, early warning dissemination	Ward-level DRM teams
NGOs/CSOs	Community mobilization, gender integration, training	Partnership frameworks
Private Sector	Resource mobilization, CSR initiatives	Public–Private Partnership Desk
County Treasury	Budget allocation and auditing	County Fiscal Strategy meetings
County Assembly	Oversight, legislation, public accountability	Policy review sessions

5.2 Stakeholder Engagement

Stakeholder participation forms the cornerstone of successful climate risk preparedness. The County Government recognizes that disaster management is a shared responsibility requiring collaboration across multiple sectors and social groups.

To institutionalize stakeholder participation, the county will establish a Multi-Stakeholder Coordination Platform (MSCP) under the County Climate Change Steering Committee consistent with the Kwale County Public Participation Policy. This platform will include representatives from government departments, NGOs, research institutions, faith-based organizations, private sector players, community-based organizations (CBOs), and special interest groups such as women, youth, and PADs.

Regular consultation forums, joint planning sessions, and feedback mechanisms will ensure inclusivity and transparency. At the ward level, participatory planning will continue through the WCCPCs, supported by community facilitators and technical officers. The county will also strengthen linkages with regional networks such as the Coastal Counties Climate Change Network (CCCCN) and the Kenya Red Cross Coastal Branch to share data, best practices, and resources.

Special attention will be given to gender-responsive engagement, ensuring that women's voices influence decision-making on resource allocation, emergency response, and recovery priorities. Youth groups will be empowered to take leadership in early warning dissemination, digital mapping, and community awareness campaigns. PADs will be provided with accessible communication channels and infrastructure to ensure meaningful inclusion. Partnership agreements will be formalized with development partners to enhance coordination, avoid duplication, and align investments with county priorities.

Through this approach, Kwale County aims to foster a participatory, transparent, and accountable disaster management ecosystem that reflects the collective strength of its people and institutions.

6 Resource Allocation

6.1 Budget Framework

The successful implementation of the Community Risk Preparedness Plan demands adequate, predictable, and well-coordinated financing mechanisms (Table 6). The County Government of Kwale, through its Treasury and the Department of Environment and Climate Change, will allocate dedicated budget lines for disaster risk management and climate resilience under the County Integrated Development Plan (CIDP 2023–2027) and the Medium-Term Expenditure Framework (MTEF).

Funding will be drawn from multiple sources, including:

- County Government Budget Allocations – Annual appropriations for climate and disaster management activities under relevant departments (Water, Agriculture, Health, Infrastructure, and Environment).
- National Government Transfers – Grants and technical assistance from institutions such as the National Treasury, NDMA, and the Climate Change Directorate.
- Development Partner Support – Contributions from organizations such as UNDP, FAO, Plan International, WWF-Kenya, and the World Bank.
- Private Sector and CSR Investments – Support from companies operating in Kwale County to fund local resilience projects.
- Community Contributions – Labour, materials, and in-kind inputs from CBOs, BMUs, and women and youth groups participating in restoration and preparedness initiatives.

Table 6: Proposed indicative budget allocation for key components of the plan

Component	Key Activities	Estimated Budget (KES Million)	Funding Sources
Prevention & Mitigation	Drainage works, water harvesting, mangrove restoration, afforestation	520	County Government, Development Partners, NGOs, Private Sector
Preparedness	Early warning systems, community training, contingency planning, stockpiling	430	NDMA/ NGOs like Plan International, County Government, Private Sector
Response	Evacuation centers, emergency supplies, rescue operations including support for SRHR	350	County Emergency Fund/Development Partner
Recovery	Infrastructure rehabilitation, livelihood support, PDNA	480	County Treasury, World Bank (Development Partners/Private Sector
Coordination & Monitoring	Capacity building, data management, stakeholder coordination	220	County Government, Development Partners, Private Sector

6.2 Resource Mapping and Logistics

Resource mapping ensures that human, financial, and material capacities are identified, mobilized, and strategically positioned before disasters occur. The County Disaster Management Unit (CDMU) will maintain an Integrated Resource Inventory (IRI) capturing available personnel, equipment, and facilities across the four sub-counties.

Key logistics and assets to be mapped include:

- Emergency shelters (schools, churches, social halls).
- Water bowsers, tractors, and mobile health units.
- Firefighting equipment and first-aid kits.
- Communication tools such as VHF radios and community public address systems.
- Transport assets, including 4x4 vehicles and boats for coastal rescue operations.

Each sub-county will designate a Resource Focal Point Officer responsible for maintaining up-to-date inventories and ensuring readiness.

Furthermore, the county will partner with local enterprises and cooperatives for procurement and storage of emergency supplies such as food, medicine, tents, and non-food items (NFIs). Gender-sensitive supply chains will be adopted to ensure that sanitary and dignity kits for women and girls are prioritized.

6.3 Sustainability and Financing Mechanisms

To ensure long-term sustainability, the County Government will operationalize the County Climate Change Fund (CCCF), enabling wards to access resilience grants for localized projects. This mechanism will be aligned with the Public Finance Management (Climate Change Fund) Regulations, 2023, ensuring compliance with transparency and equity principles.

Each ward will receive an allocation of not less than 1% of the County Annual Budget for climate action. The WCCPCs will prepare project proposals, such as water pan construction, mangrove restoration, and capacity-building workshops, for funding under this framework.

The county will also explore innovative financing instruments such as climate insurance, green bonds, and public–private partnerships (PPPs). Development partners will be encouraged to co-finance projects that integrate disaster preparedness with livelihood improvement, particularly for women and youth entrepreneurs.

A dedicated Resource Mobilization and Partnerships Unit (RMPU) will be established within the Department of Environment to coordinate donor relations, track funding flows, and facilitate reporting to the County Assembly and the Climate Change Directorate.

Through this multi-source, multi-level approach, Kwale County will maintain financial readiness for disaster preparedness, ensuring that funds, materials, and human resources are efficiently mobilized and deployed when needed.

7 Communication Plan

7.1 Internal Communication

Effective internal communication ensures that all departments, agencies, and community structures involved in disaster preparedness and response operate under a unified command system. The Kwale County Government will adopt a multi-tiered communication protocol linking the County Disaster Management Unit (CDMU), Sub-county Administrators, Ward Climate Change Planning Committees (WCCPCs), and Village Disaster Response Teams (VDRTs).

At the county level, the County Disaster Command Centre (CDCC) will serve as the primary hub for internal communication during both preparedness and emergency phases. The CDCC will operate under a 24-hour coordination framework and will utilize integrated information systems for data collection, analysis, and dissemination. The command centre will maintain direct communication with the Kenya Meteorological Department (KMD) and the National Drought Management Authority (NDMA) to receive hazard forecasts and early warning updates.

Upon receipt of risk alerts, the CDMU will immediately relay information to Sub-county Administrators through official communication channels (email, SMS, radio calls, and WhatsApp coordination groups). Sub-county Administrators will then activate Ward Administrators and local committees for dissemination to communities.

An Incident Command System (ICS) will be institutionalized to ensure clarity in command hierarchy and information flow during emergencies. This system will define communication chains, designate spokespersons, and outline reporting responsibilities at each level of response. Only authorized officers will issue official updates to minimize misinformation and panic.

To ensure real-time information flow, each ward will have an Information Officer responsible for compiling situation reports (SITREPs) and transmitting them to the county office. The use of Geographic Information Systems (GIS) and mobile data collection tools (such as KoboToolbox) will support mapping of affected areas and damage assessments.

Periodic coordination meetings between the CDMU, sectoral departments, and humanitarian partners will be conducted to review preparedness progress and communication efficiency. A digital repository will be maintained to store all records, hazard bulletins, and reports for reference and audit purposes.

7.2 External Communication

External communication aims to ensure that timely, accurate, and accessible information reaches the public, the media, and development partners before, during, and after disaster events. The Kwale County Government recognizes that transparent and inclusive communication enhances public trust, encourages early action, and supports coordinated response.

The primary mode of public communication will include community radio broadcasts (e.g., Kaya FM, Radio Kaya, and Pwani FM), public address systems, SMS alert services, and social media platforms managed by the County Information Office. During emergencies, the CDMU will issue official press releases and situation updates through the County Director of Communications.

Community barazas (public meetings) will remain the main channel for information sharing in rural areas, especially for populations with limited digital access. Chiefs and village elders will

play critical roles in mobilizing communities for early warning briefings and response coordination.

To ensure inclusivity, all communication materials will be disseminated in Kiswahili and local dialects such as Digo and Duruma, ensuring comprehension among all residents. For persons abled differently (PADs), visual aids, sign language interpreters, and radio announcements with descriptive cues will be used. Women's groups, youth associations, and persons with disabilities networks will be engaged as community communication champions.

The county will also partner with humanitarian organizations and media houses to run awareness campaigns on seasonal forecasts, flood evacuation procedures, and disease prevention. These campaigns will include radio talk shows, posters, and social media infographics developed collaboratively with NGOs like Plan International and WWF-Kenya.

To enhance feedback and accountability, the CDMU will maintain a Public Information and Feedback System (PIFS) comprising toll-free hotlines, suggestion boxes at ward offices, and online reporting portals. This system will allow residents to report hazards, share local observations, and seek assistance.

A critical component of the external communication framework is trust-building. To counter misinformation during crises, all public communication will be verified and authorized by the County Disaster Communications Desk before release. The county will collaborate with media outlets to ensure consistency and accuracy in messaging.

Table 7: Communication Channels for Disaster Preparedness and Response

Communication Channel	Target Audience	Responsible Entity	Frequency	Language/Accessibility
Community Radio (Kaya FM, Pwani FM)	General public	County Information Office, KMD	Weekly during normal periods; daily during hazards	Kiswahili & Local Dialects
SMS Alert System	Households, Traders, Fisherfolk	NDMA, CDMU	Real-time	Universal
Barazas and Public Meetings	Local communities	Chiefs, Ward Admins	Monthly or as needed	Verbal/Visual
Social Media Platforms	Youth, Urban populations	County ICT Office	Continuous	English & Kiswahili
Print Media & Posters	Schools, Institutions	County Education Dept	Seasonal	Accessible (visual)
Toll-Free Hotlines	Affected households	CDMU	24/7	Voice and text
Sign Language & Visual Aids	PADs	County Gender & Social Services	Continuous	Inclusive

8 Monitoring and Review

8.1 Drills and Simulations

Regular drills and simulations are critical to ensuring the effectiveness of the county's preparedness and response systems. Kwale County will institutionalize bi-annual multi-hazard simulation exercises at county, sub-county, and ward levels to test the operational readiness of stakeholders.

These drills will simulate events such as drought emergencies, flood evacuations, fire outbreaks, strong winds and oil spills along the coastline. The County Disaster Management Unit (CDMU), in collaboration with the National Disaster Operations Centre (NDOC) and humanitarian partners like the Kenya Red Cross Society (KRCS), will design and oversee these exercises.

Simulation activities will include early warning dissemination tests, mock evacuation of households, deployment of first responders, and activation of communication chains. Each event will conclude with after-action reviews (AARs) to identify strengths, weaknesses, and lessons learned for improvement.

Gender and social inclusion will be integral to these exercises. Women, youth, and persons abled differently (PADs) will actively participate in designing and testing evacuation routes, ensuring that facilities meet accessibility and safety standards. Female first responders will receive specialized training in gender-sensitive humanitarian action, including protection against gender-based violence (GBV) during emergencies.

In addition, schools will participate in School Safety Drills, coordinated by the County Education Department, to enhance preparedness among learners and teachers. These exercises will include fire response training, assembly point designation, and first-aid demonstrations.

8.2 Plan Updates and Reviews

Monitoring and reviewing the preparedness plan ensures that it remains relevant, effective, and responsive to changing climatic, socio-economic, and institutional contexts. The Kwale County Community Risk Preparedness Plan will undergo annual reviews and a comprehensive mid-term evaluation (in 2027) to assess progress and identify emerging risks.

The County Climate Change Steering Committee (CCCSC), chaired by the CECM for Environment, will oversee these reviews. Each implementing department will be required to submit quarterly and annual progress reports detailing achievements, challenges, and lessons learned. These reports will feed into a consolidated county preparedness report submitted to the Governor and County Assembly for oversight.

Periodic updates will also integrate new data from the Kenya Meteorological Department (KMD), National Drought Management Authority (NDMA), and global climate monitoring platforms such as IPCC projections to adjust interventions based on the latest climate scenarios.

Reviews will focus on:

- The performance of early warning systems (timeliness, accuracy, dissemination).
- Institutional responsiveness and coordination efficiency.
- Budget utilization and financial accountability.
- Gender and inclusion indicators, including participation rates of women and PADs.
- Effectiveness of recovery and rehabilitation programs.

The review outcomes will inform subsequent budget cycles, policy formulation, and integration into the County Integrated Development Plan (CIDP) and County Annual Development Plans (CADPs).

8.2.1 Performance Evaluation Framework

A robust performance evaluation framework will be used to monitor progress toward achieving the plan's objectives. Key performance indicators (KPIs) will be aligned with the county's monitoring and evaluation (M&E) system and the National Performance Contracting framework (Table 8).

Each department and stakeholder will report on a standardized Resilience Performance Matrix, covering thematic areas of prevention, preparedness, response, and recovery. The M&E framework will include both process indicators (e.g., number of trainings conducted, equipment procured) and outcome indicators (e.g., reduction in response time, improved livelihood resilience).

Table 8: Monitoring and Evaluation Framework for the Preparedness Plan (2025–2030)

Focus Area	Indicator	Responsible Institution	Reporting Frequency	Data Source
Early Warning & Awareness	% of population receiving timely alerts	NDMA, CDMU, KMD, KARLO	Quarterly	Early Warning Reports
Infrastructure Resilience	Number of flood-resilient structures constructed	Infrastructure Dept	Annual	Progress Reports
Gender Inclusion	% of women and PADs in preparedness committees	Gender Dept, CDMU	Semi-annual	Committee Registers
Livelihood Diversification	% of households adopting climate-smart livelihoods	Agriculture Dept	Annual	Household Surveys
Disaster Response Time	Average hours from alert to first response	CDMU, KRCS	Quarterly	Incident Logs
Financial Accountability	% of allocated funds utilized as per budget	Treasury Dept	Annual	Budget Review Reports

8.2.2 Learning and Knowledge Management

Kwale County recognizes that effective preparedness requires continuous learning and adaptation. A County Knowledge and Learning Hub (KLH) will be established within the Department of Environment to document lessons, case studies, and best practices from drills, disaster responses, and community projects.

The KLH will also maintain a digital repository for research reports, hazard maps, and training manuals. Partnerships with research institutions such as the Kenya Marine and Fisheries Research Institute (KMFRI), Pwani University, and WWF-Kenya will support evidence-based policy formulation and data-driven decision-making.

Community-to-community learning exchanges will be encouraged through field visits, forums, and climate resilience fairs where local groups can share innovations in water management, farming, and ecosystem restoration.

This emphasis on learning will ensure continuous improvement, enabling Kwale County to remain adaptive, resilient, and responsive to emerging climate and disaster risks.

9 Appendices

9.1 Maps of Hazard Prone Areas

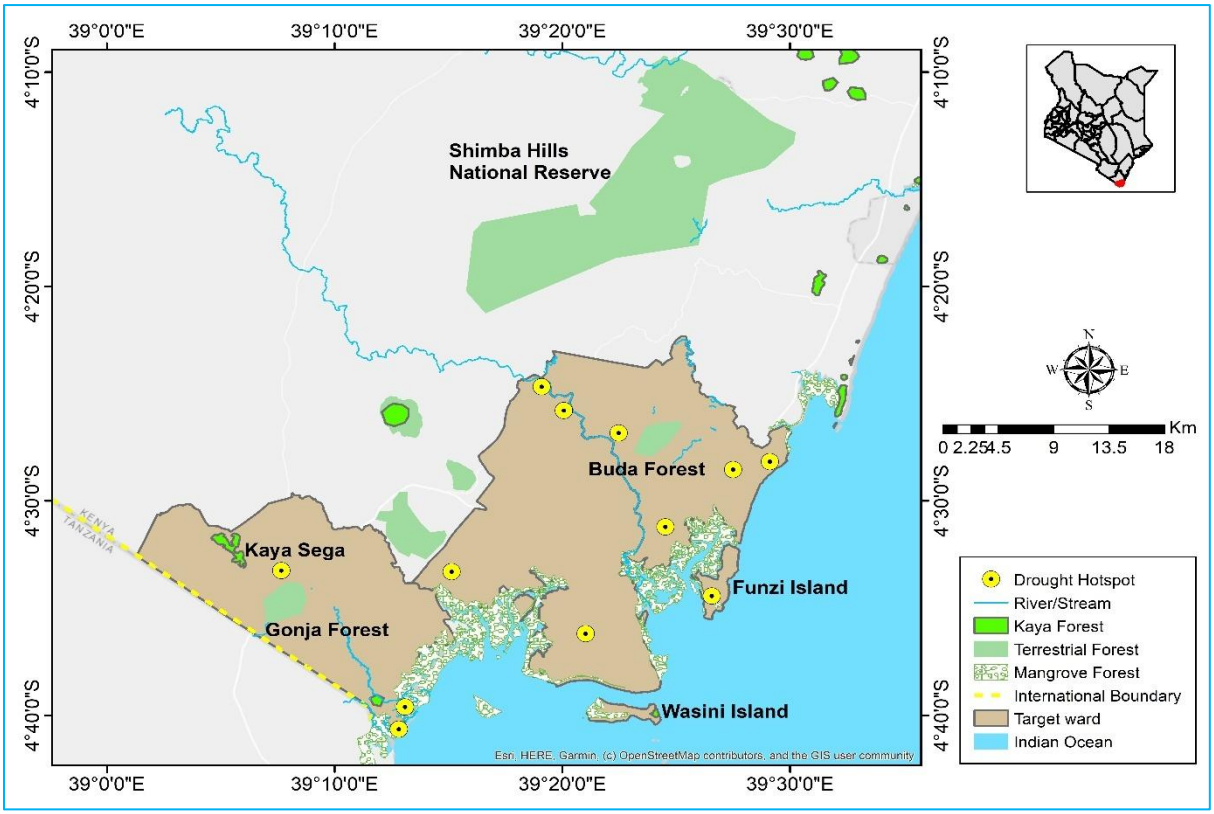

Figure 2: Drought hotspots in Kwale County (Source: PCVA, 2025)

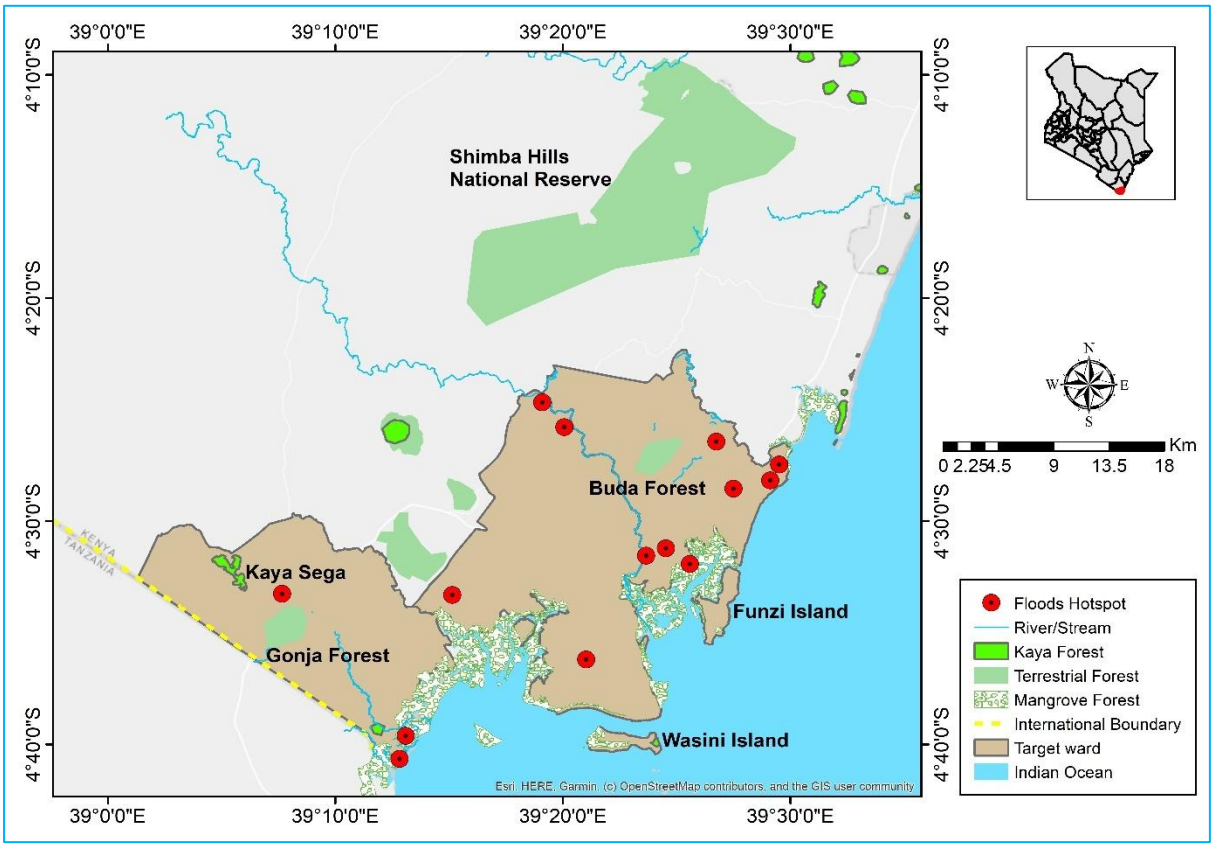

Figure 1: Flood hotspots in Kwale County (Source: PCVA, 2025)

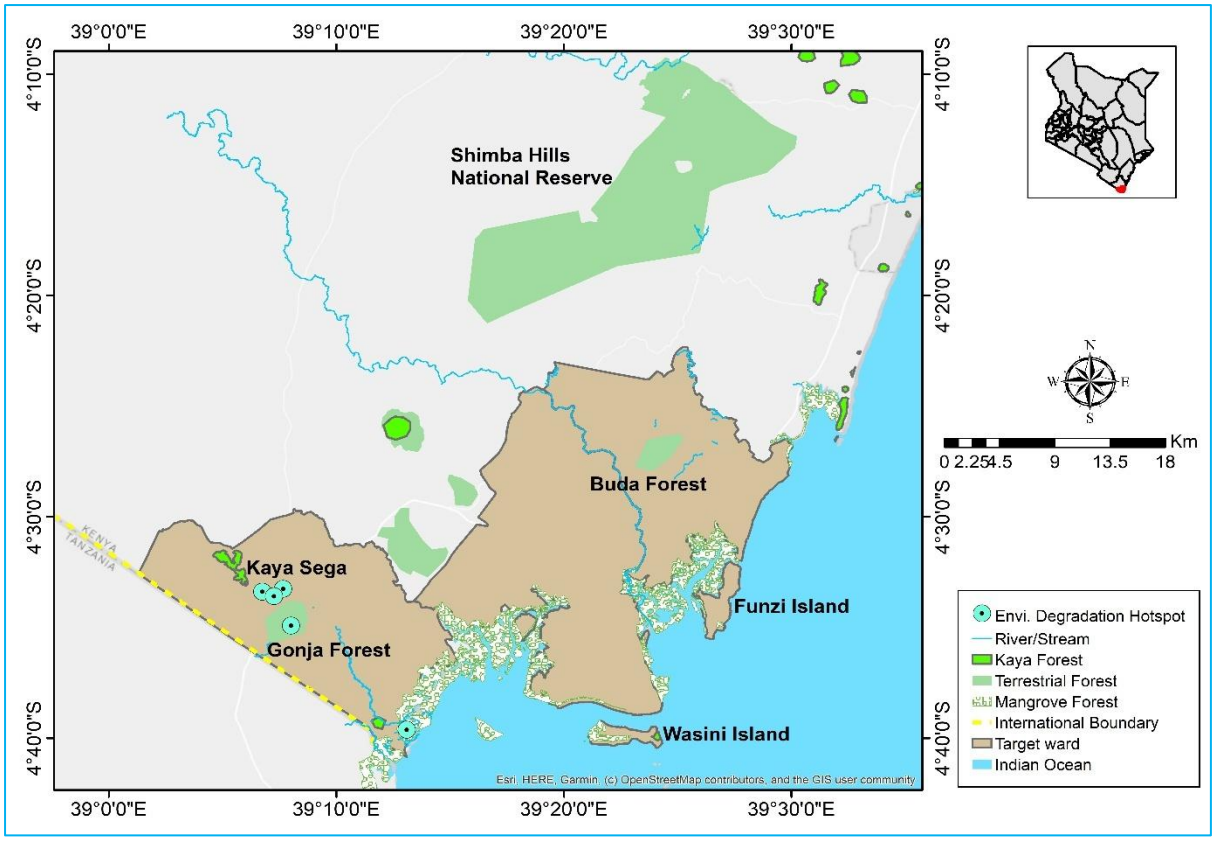

Figure 3: Environmental Degradation hotspots in Kwale County (Source: PCVA, 2025)

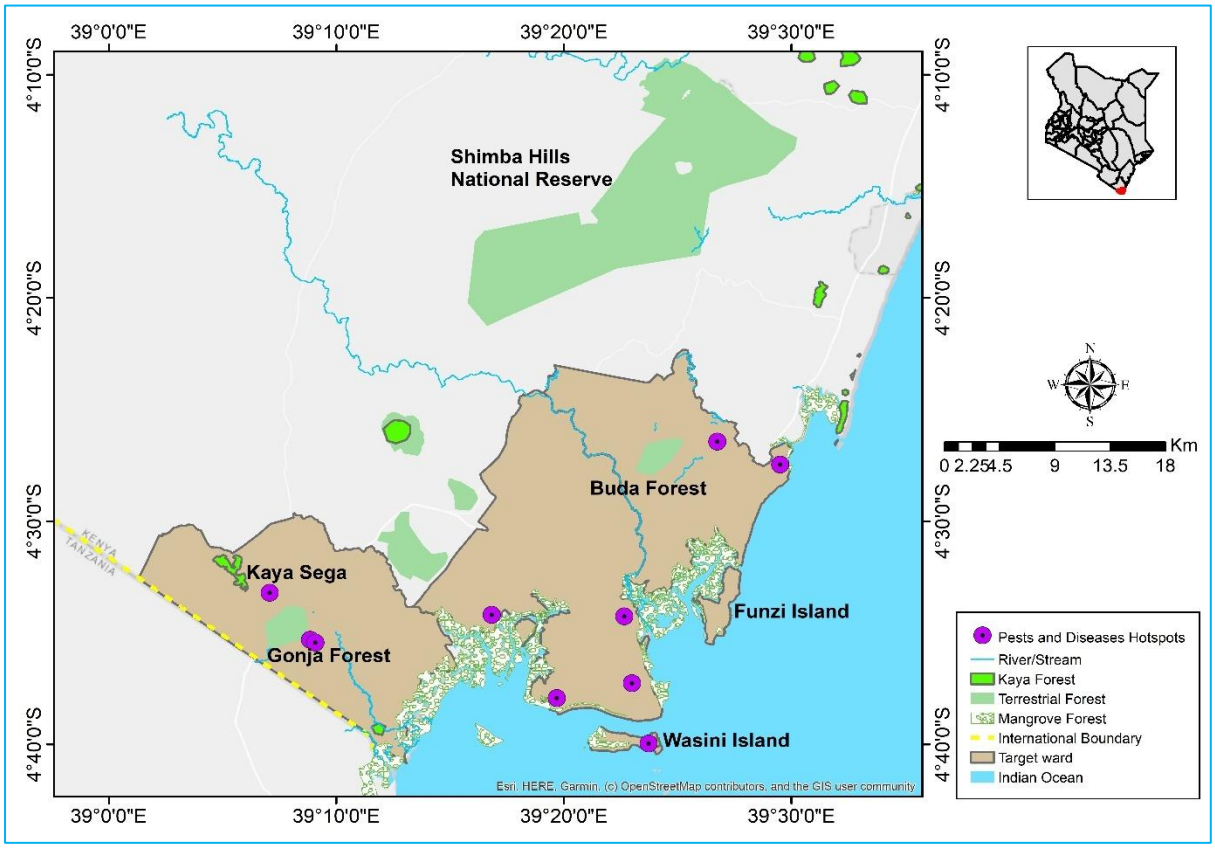

Figure 4:Pests and diseases hotspots in Kwale County (Source: PCVA,2025)

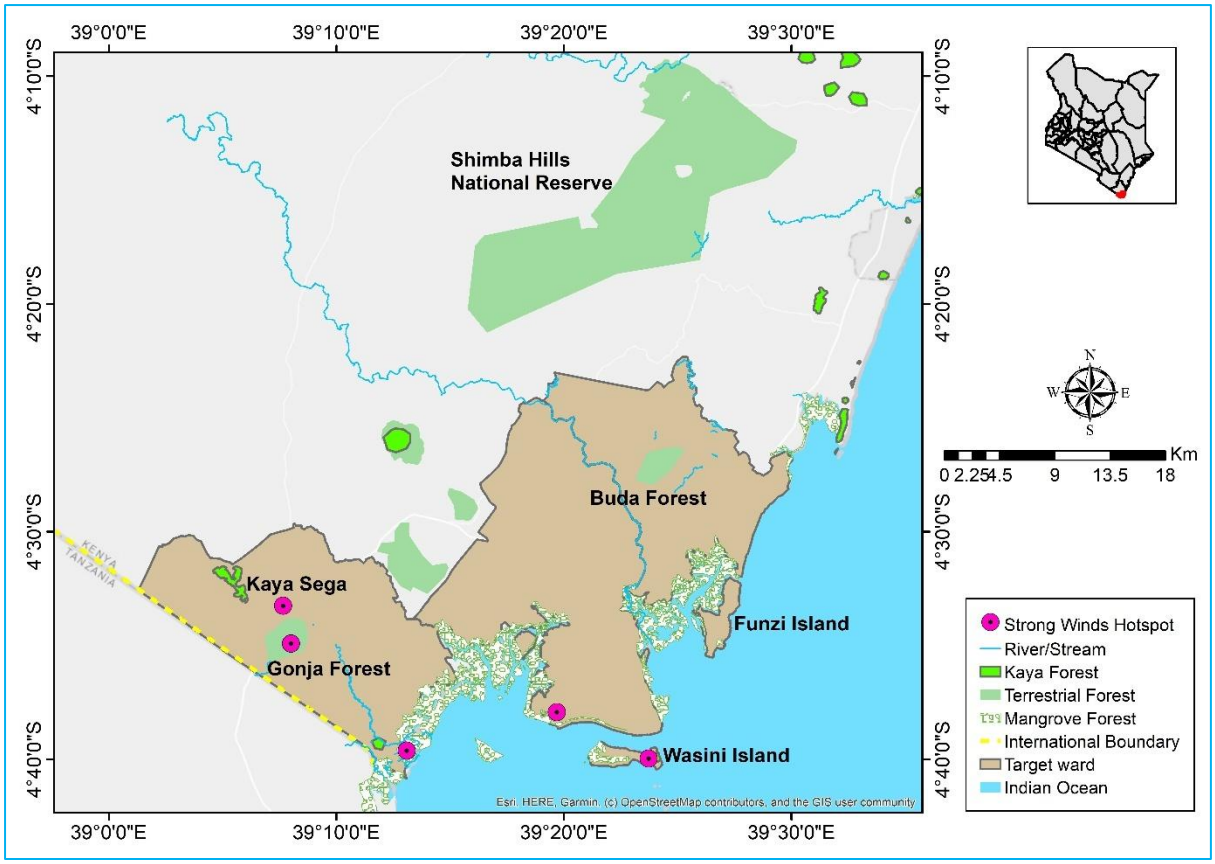

Figure 5: Strong winds hotspots in Kwale County (Source: PCVA, 2025)

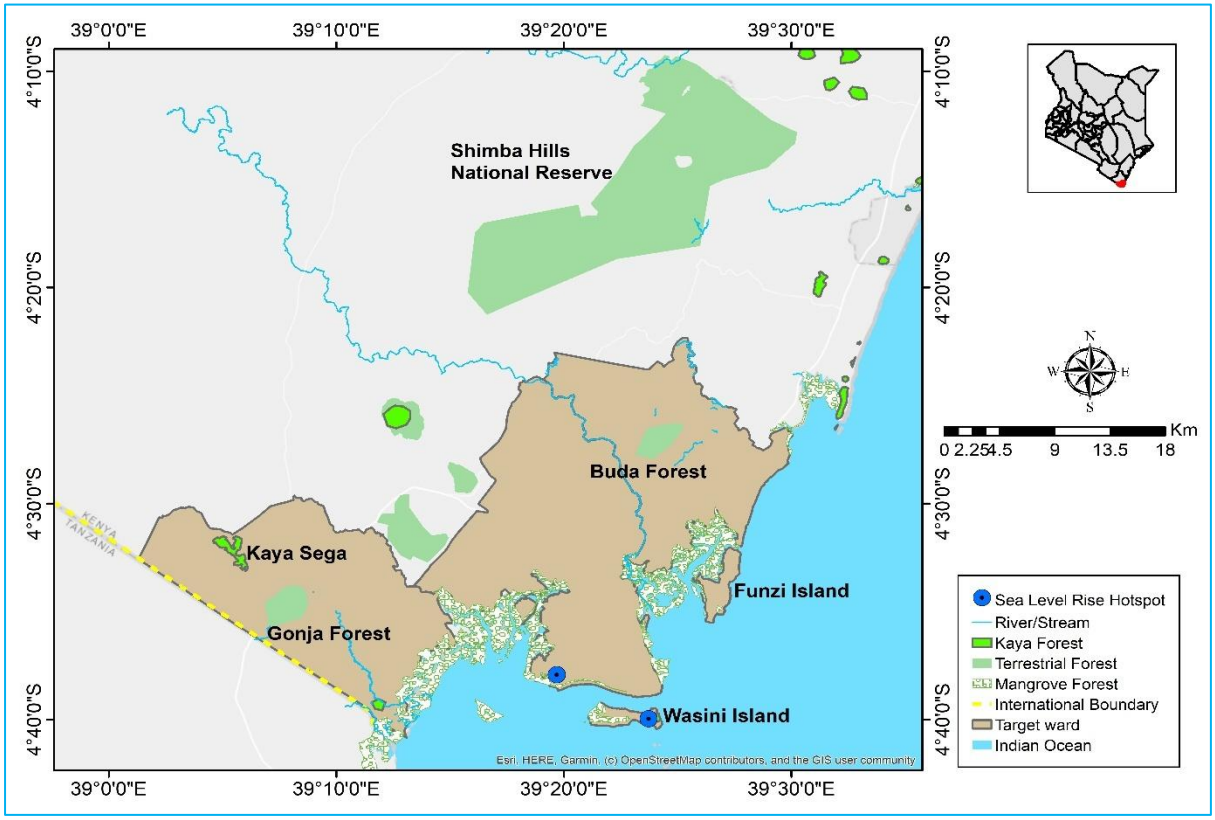

Figure 6: Sea level rise hotspots in Kwale County (Source: PCVA, 2025)

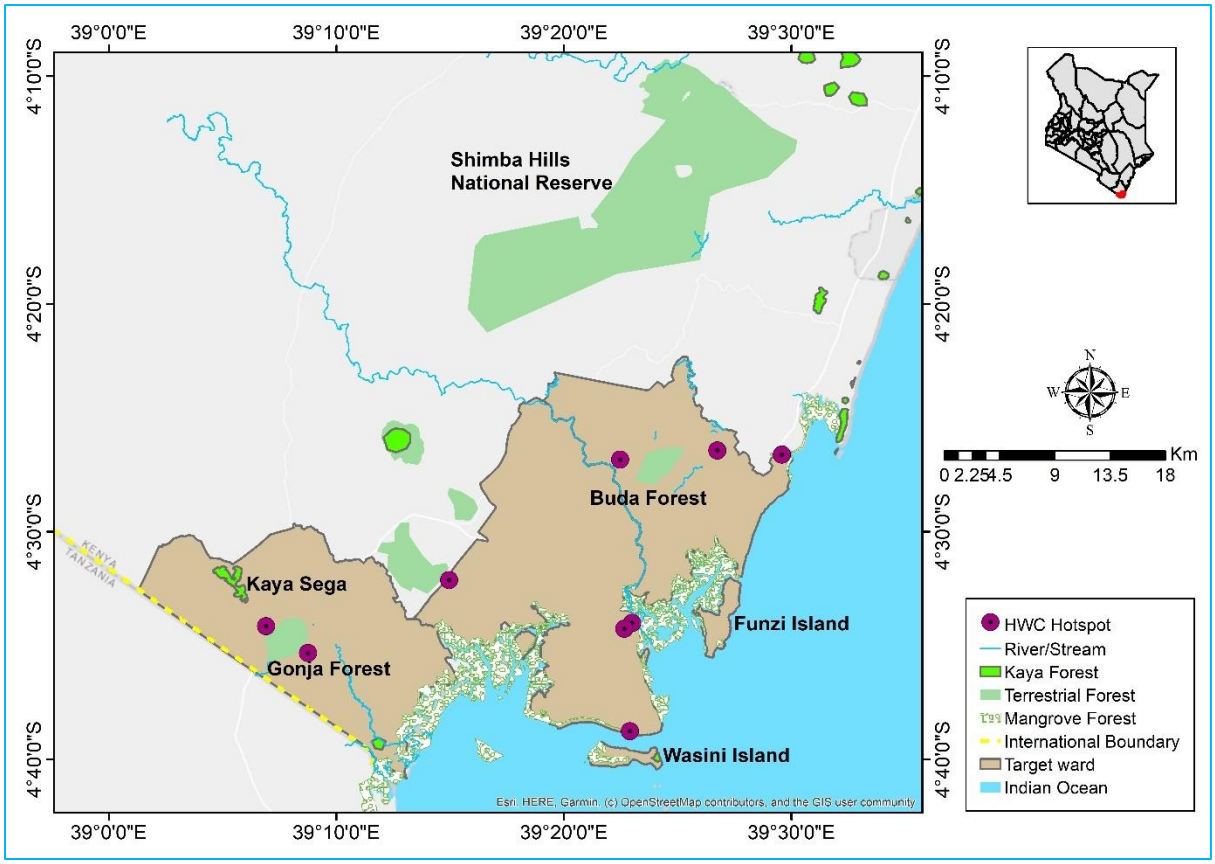

Figure 7: Human Wildlife Conflicts hotspots in Kwale County (Source: PCVA, 2025)

10 References

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